

Problem

The voltage and current at the terminals of a circuit are

$$v(t) = 128 + 192 \cos 120\pi t + 96 \cos(360\pi t - 30^\circ)$$

$$i(t) = 4 \cos(120\pi t - 10^\circ) + 1.6 \cos(360\pi t - 60^\circ)$$

Find the average power absorbed by the circuit.

Step-by-step solution

Step 1 of 2

The voltage signal is $v(t) = 128 + 192 \cos 120\pi t + 96 \cos(360\pi t - 30^\circ)$

The current signal is $i(t) = 4 \cos(120\pi t - 10^\circ) + 1.6 \cos(360\pi t - 60^\circ)$

The average power absorbed by the circuit is calculated as,

$$P_{avg} = V_{dc} I_{dc} + \frac{1}{2} \sum_{n=1}^{\infty} V_n I_n \cos(\theta_n - \phi_n) \dots\dots (1)$$

V_{dc} is the dc voltage .

I_{dc} is the dc current.

V_n is the ac voltage for the n^{th} harmonic.

I_n is the ac current for the n^{th} harmonic.

θ_n is the angle corresponding to the voltage signal.

ϕ_n is the angle corresponding to the current signal.

Step 2 of 2

Take the voltage signal $v(t) = 128 + 192 \cos 120\pi t + 96 \cos(360\pi t - 30^\circ)$.

Here, dc voltage is 128 V , the 1st harmonic voltage is 192 V with phase 0° , the third harmonic voltage is 96 V with phase 30° .

Take the current signal $i(t) = 4 \cos(120\pi t - 10^\circ) + 1.6 \cos(360\pi t - 60^\circ)$.

Here, dc current is 0 A , the 1st harmonic current is 4 A with phase 10° , the third harmonic current is 1.6 A with phase 60° .

Substitute all these values in equation (1).

$$\begin{aligned} P_{avg} &= (128)(0) + \frac{1}{2}(192)(4) \cos[0 - 10^\circ] + \frac{1}{2}(96)(1.6) \cos[30^\circ - 60^\circ] \\ &= 0 + 384 \cos(-10^\circ) + 76.8 \cos(-30^\circ) \\ &= 384 \cos(10^\circ) + 76.8 \cos(30^\circ) \\ &= (384)(0.984807753) + (76.8)(0.8660254038) \\ &= 378.1661772 + 66.51075101 \\ &= 444.6769282 \text{ W} \end{aligned}$$

The average power absorbed by the circuit is 444.6769282 W.